Explainable Shipment ETA Prediction and Delay Propagation

Aiqon AI/SE Intern — Round 2 Technical Challenge

Executive summary

This project implements a reproducible machine-learning pipeline that estimates final shipment delay, classifies the risk of a delay exceeding 24 hours, and examines which known-at-checkpoint features the model relies on. A synthetic dataset of 600 journeys is generated with explicit upstream/downstream delay interactions. A Random Forest regressor and classifier are evaluated on a held-out test set and compared with a median-delay baseline.

1. Problem framing and prediction checkpoint

The prediction checkpoint is defined as **after destination-port arrival but before customs clearance**. The model uses only features intended to be known at this point: route, route distance, planned transit/port/customs durations, observed departure delay, observed port-arrival delay, weather severity, port congestion, and customs complexity. Realized port-processing delay, customs delay, final arrival timestamp, and target columns are excluded from model inputs to reduce target leakage.

2. Dataset and delay propagation

The generator simulates 600 shipment journeys across five illustrative routes. Each row includes scheduled and actual timestamps, stage-level delays, external-condition indicators, and final delay. The synthetic process models port-processing delay as a function of congestion and observed port-arrival delay; customs delay depends on customs complexity, congestion, and port-processing delay. Final delay combines upstream and downstream effects plus stochastic variation. The binary risk label is final delay greater than 24 hours. These equations are designed assumptions, not estimated logistics laws.

3. Modelling and evaluation

A training-set median predictor provides a baseline. The primary regression model is a Random Forest with 350 trees and minimum leaf size 2. A separate Random Forest classifier estimates the probability of delay exceeding 24 hours. A fixed random 80/20 split is used, with stratification for classification. Regression metrics are MAE, RMSE, and R^2; classification metrics are precision, recall, F1, ROC-AUC, and confusion matrix.

Metric	Held-out result
Test records	120
Median baseline MAE	7.498 hours
Random Forest MAE	5.115 hours
Random Forest RMSE	6.342 hours
Regression R^2	0.498
Classifier precision / recall / F1	0.838 / 0.838 / 0.838
Classifier ROC-AUC	0.907

On this run, regression MAE improved relative to the training-median baseline. The classifier performance is measured on a held-out test set; because the data is synthetic, these metrics indicate internal consistency with the simulator rather than expected performance on real shipments.

4. Explainability and findings

Held-out permutation importance measures how much test MAE worsens when a feature is shuffled. It is a global model-reliance measure, not proof that a feature causes delay. The top features in this run are shown below.

Feature	MAE increase after shuffling (hours)
port_arrival_delay_hours	2.691 ± 0.414
port_congestion	1.229 ± 0.182
customs_complexity	0.376 ± 0.123
departure_delay_hours	0.083 ± 0.067
weather_severity	0.037 ± 0.046
planned_customs_hours	0.026 ± 0.035

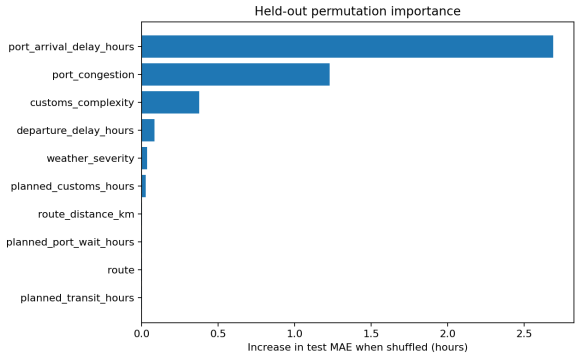


Figure 1. Held-out permutation importance.

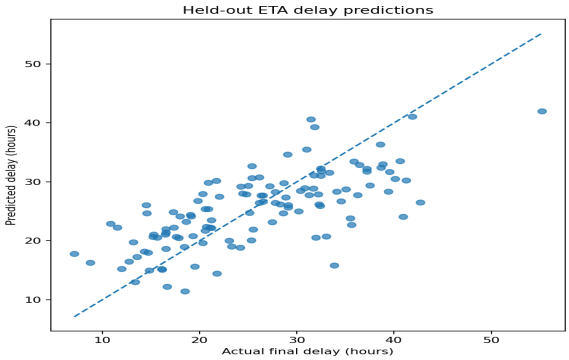


Figure 2. Actual versus predicted final delay on held-out records.

5. Implementation and reproducibility

The repository contains separate modules for data generation, feature definitions/leakage validation, model training/evaluation, explanation plots, automated tests, and an optional Streamlit demo. Run commands from the repository root: `pip install -r requirements.txt`, `python src/generate_data.py`, `python src/train_model.py`, `python src/explainability.py`, `pytest -q`, and `streamlit run app.py`.

6. Limitations and future work

The dataset is synthetic, small, and generated from assumptions authored for this challenge; it cannot establish real-world generalization. A random split does not test temporal drift. Feature importance is not causal inference. The demo does not consume live AIS, weather, port, or customs data and does not produce calibrated uncertainty intervals. Next steps are to validate against real historical timestamps, compare chronological and route-held-out splits, calibrate probabilities, add prediction intervals, monitor data drift, and obtain logistics-domain review.

Conclusion

The prototype connects stage-aware synthetic data, a defined prediction checkpoint, leakage-aware feature selection, regression and classification, held-out evaluation, and interpretable model diagnostics. Its main contribution is a transparent end-to-end structure that can be validated and extended when real shipment data becomes available.